

COMPACT-OBJECT CRUST MHD EQUIVALENCE IN HQIV: RESISTIVE REDUCTION, GEOMETRY-FIRST MULTIPOLES, AND HONEST SCOPE

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ABSTRACT. Position. Companion to the compact-object witness data bundle [2] and the O-Maxwell fluid chart [3]. This note does **not** propose a rival magnetic theory. It records that standard **resistive and Hall MHD** crust dynamics reduce onto existing HQIV slots in [Hqiv/Physics/CompactObjectMhdEquivalenceScaffold.lean](#) with explicit coefficient identification, then states where **geometry-first** organization (-tipping layers, spin-axis latitude belts, τ_{mis} obliquity) yields predictions that traditional Hall-MHD fits often bolt on afterward.

Reduction (proved naming). Traditional ohmic $E = J/\sigma$ is `ohmicAxialField`; the classical limit `coronalEffectiveAxialField\classical\_limit` is machine-checked. Resistive induction uses `inductionResistivityEta`(ξ, ϵ) = $\gamma_{\text{release}}(\xi) G_{\text{eff}}(\epsilon)$ and scalar discharges `steadyInductionFieldLt`, `inductionGrowthRateFromLt`. Turbulent viscosity split `nuTotal\_eq\_nuMol\_add\_hqivEddy` maps the traditional $\tau_{\text{mol}} + \tau_{\text{eddy}}$ bookkeeping.

Geometry-first predictions (witnessed in Python). Layered zones, colatitude channels (ρ_{pol} , Coriolis gate), τ_{mis} $m=1$ spots, and NICER J0030/J0740 overlays are generated from [scripts/hqiv_compact_object_mass.py](#) (`--surface-multipole-audit`, `--paper-dynamics-outline`).

Honest limitations. Proto-neutron-star $\alpha\Omega$ seed fields for magnetar 10^{14} – 10^{15} G are **boundary data**, not outputs of the cold-crust witness. Full vector Hall-MHD integration over 10^3 – 10^6 yr and numeric $\sigma(T, B)$ calibration remain milestones (§5).

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as `Fin 4`, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.

Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation ($Q_L, u^c, d^c, L, e^c, \nu^c$), the patch trace identities give zero $U(1)_Y^3$, gravitational- $U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite `Fin 3` label sum of the same cancelled one-generation block, not a fitted multiplicity.
- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ -independent; and abelian commutator obstructions vanish.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $G_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour

commutator certificates are discharged on the octonion carrier (see the sector zeta and holonomy discharge bundle in the Lean catalog). ****Matrix Wilson layer:**** full four-edge $SO(8)$ generator transport and Hopf-shell T^4 torsion plaquettes with pinned non-abelian witness ([Hqiv/Physics/S08PlaquetteHolonomy.lean](#), witness `so8PlaquetteHolonomyDischarged_holds`). ****Wilson-kinetic equivalence**** ([Hqiv/Physics/ActionHolonomyGlue.lean](#), witness `wilsonKineticPlaquetteEquivalence_discharged`). ****Partial / open:**** Haar measure on rotated charts ([Hqiv/Physics/NonAbelianHolonomyMeasureScaffold.lean](#)); general flat HQVM kinetic boost and full strong-interacting discrete-action Poincaré ([Hqiv/Physics/DiscreteActionStrongPoincareBridge.lean](#), [Hqiv/Physics/DiscreteActionPoincareScaffold.lean](#)). Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart.

- **Rapidity phase and Lorentz invariance.** The null-lattice carrier `NullLatticeEvent` charts into 1+1 Minkowski space by the classical rapidity parametrization $\Lambda(\eta)(1, 1)$; rapidity addition on the carrier is equivariant under `boostMatrix11`, the Minkowski pairing is bilinearly invariant, and the polar-angle / `zetaHQIVTerm` phase is unchanged when ϕ transforms as a Lorentz scalar. The honest 3+1 extension preserves `minkowskiSq4` on the embedded $(t, x^1, 0, 0)$ plane.
- **Spatial rotations and directional readouts.** Orthogonal $O(3)$ rotations on `Fin 3` spatial vectors preserve Euclidean inner products, $\|\mathbf{r} \times \mathbf{v}\|^2$, axis-projection fractions, and the spatial block of `minkowskiSq4` when time is fixed ([Hqiv/Geometry/SpatialRotationLorentzClosure.lean](#), witness `spatial_rotation_lorentz_closure_discharged`). Applied discharge: CMB sky rotation shifts the Stokes angle additively while preserving $\cos^2 + \sin^2 = 1$; O-Maxwell fluid sources rotate covariantly with gradients; coronal axis projections are joint rotation invariants. Full patch Lorentz on the flat chart is the rapidity bundle plus this spatial bundle (witness `full_lorentz_closure_discharged`). Continuum interacting QFT / Wightman theorems are not claimed.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and sector zeta bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

Projection geometry and the classical ζ dictionary (2026-06). HQIV uses *zeta* in two distinct, now formally separated roles:

- **Sector zeta on the patch carrier** (TUFT+SM synthesis): fibre-torsion subleading bodies, `zetaHQIVTerm` phase on Hopf shells, and holonomy discharge bundles on the octonion light-cone carrier—finite spectral readouts indexed by shell and Fano sector, not the Riemann function.
- **Classical $\zeta(s)$ as an $SO(4)$ projection readout** ([1]): the Mathlib Riemann zeta packaged as a *regional* closed form on $\mathbb{C}$ —Dirichlet/Euler for $\Re s > 1$, $\sin\text{--}\cos\text{--}\Gamma$ assembly on the strip, Bernoulli slots—with the functional-equation pair $(\sigma, 1 - \sigma)$ interpreted as a 45° quaternionic projection. Machine-checked on this spine: harmonic divergence forces Δ and $\mathfrak{so}(4)$ closure; shell sum $= \zeta(s)$ for $\Re s > 1$; polar decoupling $n^{-s} = n^{-\sigma} \cdot e^{-it \log n}$ (modulus σ -only, phase t -only); prime-log Diophantine independence; Goldbach circle $pq + (q - N)^2 = N^2$ as additive curvature in the log channel; and a large family of RH-*equivalent* locators (twiddle, spectral weights, polar collapse, transformer adjoint law, S^7 torsion cancellation) with *zero slack*—not a discharge of RH or Goldbach parity.

Continuum heat-flow and de Bruijn–Newman notation in the same story spine are *comparison ports* (`dbnHeatFamily`, lattice dominance hypotheses); they are not load-bearing continuum extensions of the patch. Formal mathematical notes may include only the readout-dictionary paragraph (`readout_dictionary_messaging.tex`) rather than this full contract when the manuscript is projection geometry rather than patch phenomenology.

1. INTRODUCTION AND FRAMING

Neutron-star magnetospheres and crusts are modeled in traditional astrophysics as **composite MHD**: inhomogeneous Maxwell equations, conducting fluid momentum, and conductivity $\sigma(T, B)$ closures, often extended with Hall drift and crust elasticity [2]. HQIV records the same stack at the action level: **O-Maxwell** $S_O(J, A, \phi)$ plus **modified fluid** closure ([Hqiv/Physics/HQIVFluidClosureScaffold.lean](#)).

The claim of this paper is therefore *organizational*, not phenomenological:

We have a first-principles way to organize the same physics around -tipping layers and the spin axis, yielding testable predictions for mass/spin dependence and surface multipoles.

2. EQUATION-LEVEL EQUIVALENCE

Table 1 displays standard resistive-MHD equations beside HQIV Lean names. Coefficient identification is explicit; vector $\nabla \times$ PDEs are **scalar discharges** on the crust witness chart until the vector milestone (§5).

TABLE 1. Traditional MHD $\leftrightarrow$ HQIV slot reduction (see [Hqiv/Physics/CompactObjectMhdEquivalenceScaffold.lean](#)).

Traditional	HQIV slot / coefficient ID
Inhomogeneous Maxwell $\nabla \times F = J$	<code>EL_0_general</code> (Hqiv/Physics/Action.lean)
$E = J/\sigma$ (ohmic)	<code>ohmicAxialField</code> ; classical limit proved
$\partial_t B \sim \nabla \times (v \times B) - \nabla \times (\eta \nabla \times B)$	<code>inductionGrowthRateFromLt</code> , <code>steadyInductionFieldLt</code>
MHD resistivity η	$\eta_{\text{HQIV}}(\xi, \varepsilon) = \gamma \text{release}(\xi) G_{\text{eff}}(\varepsilon)$
$\rho(\partial_t v + v \cdot \nabla v) = \dots + (J \times B)/c + \nabla \cdot \tau$	<code>hqvFluidInertiaFactor</code> ; Maxwell stress from S_O
$\tau_{\text{mol}} + \tau_{\text{eddy}}$	<code>nuTotal_eq_nuMol_add_hqvEddy</code>
PNS turbulent α (dynamo driving)	<code>hqvEddyViscosity</code> + <code>hqvVacuumMomentumSource3</code> (F2 chart)
Hall $\partial_t B \propto \sigma/(ne) \nabla \times ((\nabla \times B) \times B)$	<code>rapidityNormalized_frozenFirstIndexJet</code> (milestone)

Proposition 2.1 (Discharged resistive identification). *For any environment slots (ξ, ε) , traditional resistivity η may be identified with `inductionResistivityEta` (ξ, ε) , which is nonnegative (`inductionResistivityEta\_nonneg`) and equals γ at lock-in with zero gravity (`inductionResistivityEta\_lockin\_zero\_gravity`). The bundle `mkCompactObjectMhdEquivalenceDischarged` witnesses the reduction.*

3. GEOMETRY-FIRST STRUCTURE

Traditional codes can impose composition layers and offset hotspots by hand. HQIV discharges the following from -lattice and spin-chart slots:

- (1) **-tipping layers.** Nuclear, charmed, and top zones with radii r_{top} , r_{charm} tied to ε ; charm retreat on spindown (`scripts/hqv_compact_object_mass.py`, `--spindown-charm-audit`).

- (2) **Spin-axis latitude belts.** Pole channel (ρ_{pol} , axial $\nabla\phi$), mid-latitude Coriolis gate $\sin^2\theta|\cos\theta|$, equatorial induction.
- (3) **Unified obliquity.** Misaligning torque `crustMisalignTorqueFromStressDivergence` vs aligning magnetic braking; enhanced closure uses lattice-weighted **B.align**, charm-ledger boost, and minimal crust $J \times B$ ([scripts/hqiv_compact_object_mass.py](#)).
- (4) **Surface multipoles.** Zonal $l = 2, l = 3$ and $m = 1$ from τ_{mis} tilted-dipole modulation ($m=1$ tilt from enhanced α_{eq}); NICER overlays for J0030+0451 and J0740+6620.
- (5) **Environment-aware η .** Outside temperature with co-spin and CMB Doppler boost feeds ξ and hence $\eta(\xi, \varepsilon)$.

The high-spin / high-mass tail shows a documented transition: moderate $\Omega/\Omega_{\text{break}}$ favors Coriolis $\times$ shear ($l_2/l_0 \sim 1$); breakup favors equatorial induction ($l_3/l_0 \sim 1$)—see [data/compact_object_witnesses.json](#) and `--surface-multipole-audit`.

4. DIFFERENTIATING PREDICTIONS

Field obliquity vs surface geometry. α_{eq} from slip-torque balance is the misalignment between spin and the dipole axis used in aligning torque—not colatitude of a hot spot. Catalog dipole B for millisecond pulsars typically yields $\alpha_{\text{eq}} \approx 90^\circ$ (misaligning-dominated); the canonical witness at 10^{12} G gives $\alpha_{\text{eq}} \approx 71^\circ$ for a $1.4 M_\odot$ ms star; breakup favors low obliquity ($\sim 11^\circ$). NICER centroid offsets and $m = 1$ structure are a **separate** surface-multipole layer (`tau_mis_m1_gate`).

- **Mass/spin $\rightarrow$ multipoles.** l_2/l_0 , l_3/l_0 , centroid offset, and m_1 fraction correlate with $\Omega/\Omega_{\text{break}}$ without per-source hotspot offsets.
- **Obliquity regimes.** Enhanced aligning torque: [data/pulsar_witness_comparison.json](#) (`alpha_equilibrium_aligning_enhanced_deg`, `alpha_equilibrium_canonical_b_deg`).
- **High-mass tail.** Charm retreat, pair margin, incremental B_{charm} in young massive rotators.
- **$\eta(\xi, \varepsilon)$ calibration.** `coefficient_calibration_witness` vs literature σ and $\tau_{\text{ohm}} \sim \mu_0\sigma R^2$ (`--eta-calibration-audit`).

5. COEFFICIENT DISCHARGE AND LIMITATIONS

Discharging coefficients (not a weakness). Traditional decades of work fix $\sigma(T, B)$, Hall coefficients, and τ_{ohm} . HQIV fixes the **algebra** ($\alpha, \gamma, \varepsilon, \xi$) and proposes

$$(1) \quad \eta_{\text{HQIV}}(\xi, \varepsilon) = \gamma \text{release}(\xi) G_{\text{eff}}(\varepsilon), \quad E_{\text{Ohm}} = J/\sigma \Leftrightarrow \text{ohmicAxialField}.$$

A pre-finalization table compares η_{HQIV} and schematic $\partial B/\partial t$ against literature crust values; the numeric witness is `coefficient_calibration_witness` (`--eta-calibration-audit`), mapping $\tau_{\text{ohm}} \approx \mu_0\sigma R^2$ at stellar radius to the 10^4 – 10^6 yr band for $\sigma \in [10^7, 10^9]$ S/m.

Not claimed.

- Magnetar 10^{14} – 10^{15} G **generation** from cold-crust induction alone; B_0 is boundary data (proto-NS dynamo or catalog).
- Replacement of mature Hall-MHD crust evolution codes.
- Full time-dependent vector multipole evolution without coupling $\partial_t c_\ell$ to induction and diffusion.
- Numeric Hall-term benchmark until `rapidityNormalized\_frozenFirstIndexJet` is discharged as an explicit vector identity.

Machine-checked honesty ledger. `CompactObjectMhdProvedReductions`, `CompactObjectMhdMilestones`, and `CompactObjectMhdNotClaimed` in [Hqiv/Physics/CompactObjectMhdEquivalenceScaffold.lean](#) separate proved reductions, milestones, and explicit non-claims.

6. DATA AND REPRODUCIBILITY

- Witness JSON: `data/compact_object_witnesses.json` (`scripts/hqiv_compact_object_mass.py--json`).
- Paper bundle: `--paper-dynamics-outline, --mhd-equivalence-audit, --eta-calibration-audit`.
- Pulsar overlay: `data/pulsar_witness_comparison.json` (`scripts/hqiv_pulsar_witness_benchmark.py--json`).
- Regenerate all: `papers/compact_object_witness/regenerate_data.sh`.
- Lagrangian audit: `scripts/hqiv_lagrangian_faithfulness_audit.py--json`.

REFERENCES

1. Steven Ettinger Jr, $\zeta(s)$ as an $SO(4)$ projection to the complex plane: Harmonic- δ closure, quaternion axis readout, and regional closed forms, Lean formal proof note in `papers/s3_zeta_so4_projection/`; build lake build HQIVStory., 2026.
2. HQIV Collaboration, *Compact-object witness bundle (curvature slots, slip torque, induction closure)*, Repository data `data/compact_object_witnesses.json`, Generated by `scripts/hqiv_compact_object_mass.py`; see `papers/compact_object_witness/`.
3. ———, *HQIV o-maxwell fluid chart (mhd composition hypothesis)*, Preprint `papers/omaxwell_fluid_chart/HQIV_OMaxwell_fluid_chart.tex`, F2 chart bundle; see `AGENTS/FLUID_OMAXWELL_ROADMAP.md`.

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